A Triple-Timescale EWMA Approach to Streaming Probabilistic Forecasting

February 2026

1 Introduction

1.1 Falcon Challenge Overview

The Falcon Challenge poses a streaming probabilistic forecasting problem in which participants must predict the forward distribution of a dove's location based on real-time observations. Unlike traditional forecasting competitions, success requires:

- **Full distributional predictions:** Point estimates are insufficient; models must output probability densities that accurately quantify uncertainty.
- **Streaming-only processing:** No batch retraining or historical lookback is permitted. Models must maintain constant memory and process observations sequentially.
- **EWMA-weighted scoring:** Performance is evaluated via an exponentially weighted moving average of log-likelihoods, with a 50/50 blend of 20-second and 20-minute timescales.
- **Wealth redistribution mechanism:** Competitor wealth is redistributed based on relative EWMA scores, creating a zero-sum competitive dynamic.

1.2 Problem Framing and Constraints

Given a stream of observations $\{x_t, t_t\}$ where x_t represents dove location at timestamp t_t, the objective is to produce at each tick a probability density function $p_t(x_{t+horizon})$ that maximizes the following blended likelihood:

$$EWMA_{\text{blend}} = 0.5 \times EWMA_{\text{short}} \left(\log p_t(x_{t+\text{horizon}})\right) + 0.5 \times EWMA_{\text{long}} \left(\log p_t(x_{t+\text{horizon}})\right) \quad (1)$$

Scoring parameters:

- $EWMA_{\text{short}}$: uses $\alpha \approx 0.003$ (20-second memory).
- $EWMA_{\text{long}}$: uses $\alpha \approx 0.00005$ (20-minute memory).

Critical constraints:

- Constant memory: $\mathcal{O}(1)$ storage regardless of stream length.
- Constant time: $\mathcal{O}(1)$ computation per tick.
- No lookahead: Predictions must be strictly causal.
- No retraining: Model parameters cannot be recomputed from scratch.

2 Problem Characteristics

2.1 Streaming Nature

Standard probabilistic models (e.g., Gaussian processes) require batch training, which the Falcon Challenge prohibits. This eliminates:

- Likelihood optimization via gradient descent.
- Maximum likelihood estimation from samples.
- Online cross-validation and batch recalibration are prohibited.
- Any form of batch recalibration.

2.2 Non-Stationarity

The dove's movement dynamics are non-stationary with multiple regime types:

- **Low-volatility regimes:** Predictable increments ($\sigma \approx 0.01$–0.03); requires tight predictive distributions.
- **High-volatility regimes:** Erratic movements ($\sigma \approx 0.10$–0.30); requires wide distributions to avoid likelihood drops.
- **Rollover periods:** Daily maintenance windows (20:55–21:35 UTC) where data quality degrades.

2.3 Regime Shifts and Rollovers

- **Regime shifts:** Endogenous changes potentially driven by falcon pursuit patterns or invisible external factors.
- **Rollovers:** Exogenous infrastructure events characterized by data instability and unreliable scores.

3 Design Philosophy

3.1 Robustness vs. Sensitivity

EWMA scoring creates a tradeoff:

- High sensitivity: Rapidly tracks changes but is vulnerable to noise.
- High robustness: Stable during consistent periods but lags behind regime changes.

Resolution: multi-timescale decomposition–maintaining multiple variance estimators at different adaptation speeds and blending them.

3.2 Multi-Timescale Reasoning

The 50/50 blend requires simultaneous optimization of short-term (20s) and long-term (20m) components.

Resolution: multi-timescale variance decomposition–using fast, medium, and slow estimators concurrently, where the intermediate estimator ($\alpha = 0.0003$) bridges the adaptation gap.

4 Triple Timescale Variance Estimation

4.1 EWMA Intuition and Formulation

EWMA provides a streaming-compatible method for variance estimation. For changes $\{\Delta x_t\}$, variance is:

$$V_t \approx (1 - \alpha) \cdot V_{t-1} + \alpha \cdot (\Delta x_t - \mu_t)^2 \tag{2}$$

Effective window: $N_{\text{eff}} \approx 1/\alpha$.
Streaming properties: Constant memory, constant time, online, and causal.

4.2 Fast, Medium, Slow Estimators

Three estimators are instantiated:

- `ewa_fast`: $\alpha = 0.003$ (20-second memory).
- `ewa_medium`: $\alpha = 0.0003$ (3-minute memory).
- `ewa_slow`: $\alpha = 0.00005$ (20-minute memory).

4.3 Winsorization Rationale

Outliers can inflate variance for thousands of ticks. The solution clips changes before updates:

- Fast threshold: $1.5 \times \sqrt{V_{\text{fast}}}$.

- Medium threshold: $2.5 \times \sqrt{V_{\text{medium}}}$.
- Slow threshold: $3.0 \times \sqrt{V_{\text{slow}}}$.

Differential thresholds allow the fast estimator to reject noise while the slow estimator allows true changes to propagate.

5 Volatility Regime Awareness

5.1 Rolling vs. Baseline Variance

- **Rolling variance:** 20-tick sliding window providing unbiased instant volatility.
- **Baseline variance:** Captured after 500 ticks as a "typical" volatility anchor.

Volatility ratio:

$$vol\_ratio = \frac{\text{rolling\_variance}}{\text{baseline\_variance}}. \tag{3}$$

5.2 Adaptive Behavior Under Volatility Changes

- High volatility ($vol\_ratio > 2.0$): Weights = [0.60, 0.30, 0.10] (Fast, Slow, Medium). Shifts to slow estimator for stability.
- Low volatility ($vol\_ratio < 0.5$): Weights = [0.80, 0.15, 0.05]. Shifts to fast estimator for precision.
- Normal regime: Weights = [0.70, 0.20, 0.10] (default).

6 Rollover Detection

6.1 Time-Based Rollover Logic

A time-based detector exploits fixed UTC maintenance windows (20:55–21:35). It provides zero-lag, deterministic detection but is brittle to schedule changes.

6.2 Likelihood-Based Rollover Logic

A likelihood-based detector monitors for sudden drops. A rollover is flagged when the current log-likelihood falls below 50% of the recent 20-tick average. This is data-driven and adaptive but carries approximately a 20-tick lag.

6.3 Impact on Predictive Uncertainty

- **Scale widening:** All scales are increased by 50% during rollovers to prevent catastrophic failure.
- **Weight rebalancing:** Fast estimator weight drops to 40% to reduce sensitivity to erratic data.

7 Predictive Distribution Construction

7.1 Laplace Distribution Choice

A Laplace predictive distribution is selected for having heavier tails than the Gaussian and being computationally simpler than the Student's t distribution. In validation experiments, this choice achieved an average log-likelihood of approximately -2.18.

7.2 Scale Derivation and Constraints

For a Laplace distribution with variance σ^2, the scale parameter is:

$$b = \frac{\sigma}{\sqrt{2}} \tag{4}$$

A minimum scale of 10^{-5} is enforced to prevent numerical underflow.
Ordering constraints ensure component diversity:

$$medium\_scale \geq 1.3 \times fast\_scale \tag{5}$$
$$slow\_scale \geq 1.8 \times fast\_scale \tag{6}$$

8 Strengths and Limitations

8.1 Strengths

- True $\mathcal{O}(1)$ streaming compatibility.
- Directly targets scoring function timescales.
- Outlier robustness via winsorization.

8.2 Limitations

- Distributional rigidity: cannot capture multimodality.
- Fixed timescales may be suboptimal for other blends.
- Location-only mixture ignores directional bias or momentum.

9 Conclusion

The TripleTimescale tracker resolves the stability–reactivity tradeoff through multi-timescale decomposition and adaptive mixture weighting. It prioritizes robustness and theoretical soundness, targeting the specific EWMA scoring mechanics of the Falcon Challenge.